Finite State Automata in Natural Language Processing

DSA0302 – Natural Language Processing

Presented by:
S B Sabesh - 192472153

Guided By:
Dr.T.Devi

Introduction

- **Overview of the Project:**
- Finite State Automata (FSA) is a mathematical model used to recognize patterns and process input strings.
- It consists of states, transitions, a start state, and one or more final (accepting) states.
- In Natural Language Processing (NLP), FSA is used to recognize words, tokens, and valid language patterns.
- FSA plays an important role in lexical analysis, language recognition, spell checking, and text processing.

Deterministic Finite Automata (DFA)

- Deterministic Finite Automata (DFA) is a finite automaton in which each state has exactly one transition for every input symbol.
- It reads one character at a time and moves from one state to another until the input ends.
- It is fast and efficient, making it widely used in lexical analysis and compiler design.
- If the final state is reached, the string is accepted; otherwise, it is rejected.

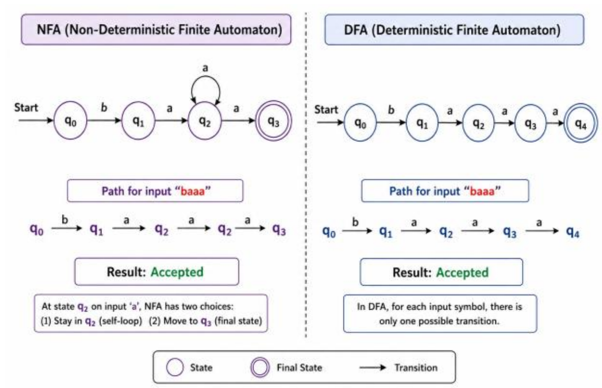
Non-Deterministic Finite Automata (NFA)

- Non-Deterministic Finite Automata (NFA) is a finite automaton where a state can have more than one transition for the same input symbol.
- NFA can follow multiple paths at the same time while processing an input string.
- NFA is mainly used in regular expression matching, pattern recognition, and Natural Language Processing (NLP).
- **Example:** For the input **"baaa"**, NFA may follow multiple paths, but if one path reaches the final state, the string is **accepted**.

Applications of FSA in Natural Language Processing

- **Lexical Analysis** – Breaks text into words or tokens for easy processing.
- **Spell Checking** – Detects incorrect words and suggests corrections.
- **Pattern Matching** – Finds specific words or text patterns in documents.
- **Language Recognition** – Checks whether the input follows the rules of a language.
- **Chatbots and Virtual Assistants** – Helps understand user input and generate appropriate responses.
- **Speech Recognition** – Converts spoken words into text by recognizing language patterns.

EXAMPLE



CONCLUSION

- DFA processes the input using one fixed path, making it fast and easy to implement.
- NFA can follow multiple possible paths, making it more flexible for pattern matching.
- Both DFA and NFA recognize valid input strings and produce the same final result.
- In Natural Language Processing (NLP), Finite State Automata improve lexical analysis, token recognition, and language processing.